

Accelerating Scoping Reviews: A Case Study in the User-Centered Design of an AI-Enabled Interdisciplinary Research Tool

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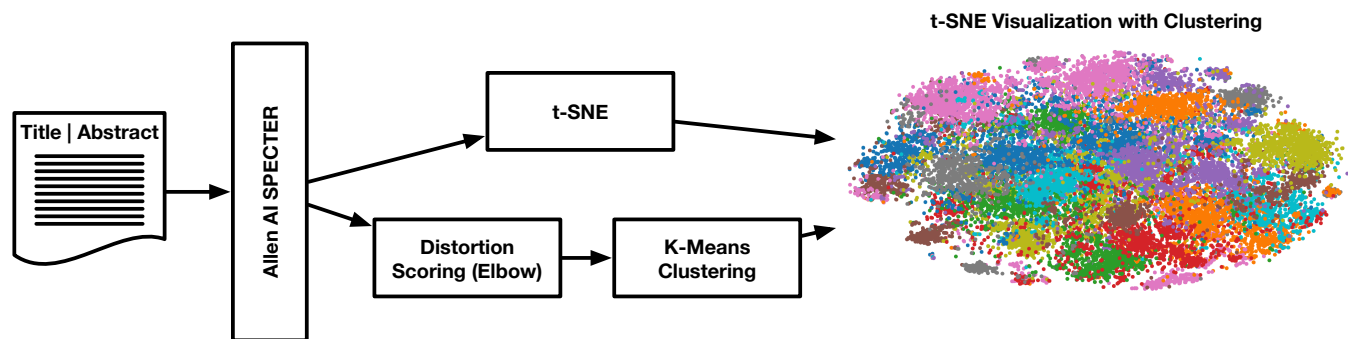


Figure 1: Approach overview. All papers’ titles and abstracts are processed through SPECTER to derive high-dimensional semantic embeddings. Data is visualized with a relationship-preserving two-dimensional representation via t-SNE. High-dimensional embeddings are evaluated and clustered on a separate path. Lastly, cluster assignments are used to color each data point.

ABSTRACT

This case study presents the user-centered design process our multidisciplinary team of computer scientists, librarians and social scientists engaged in to develop a research tool utilized as part of a scoping review to explore the vast literature on virtual humans. Our process was guided by the Information Systems Research (ISR) Framework to define the parameters of this AI-enabled accessible tool: a semantically organized, interactive evidence map, clustered

by salient topics and linked to Google Scholar to expedite the discovery of relevant resources. Specifically, we aimed to achieve several desiderata: (1) replicability, (2) objectivity, (3) automation & scalability, and (4) ease of discovery. Lessons learned include 1) how to apply the ISR Framework to benefit multidisciplinary collaboration, 2) user-centered design benefits from in-house, cross-discipline training, 3) interdisciplinary challenges benefit from multidisciplinary teams, and 4) intentional feedback loops remove the pressure from brainstorming and provide equal opportunity for multidisciplinary input throughout the development life cycle. To enable replicability, we provide full access to the dataset, the AI-enabled interactive evidence map, and source code.

CCS CONCEPTS

• **Information systems** → **Content analysis and feature selection**; **Document representation**; • **Human-centered computing** → *Visualization systems and tools*.

KEYWORDS

Scoping reviews, user-centered design, representation learning, virtual human

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1 INTRODUCTION

For many, gone are the days of sifting through dusty card catalogs in a library. Those who have spent hours trying to locate a single misfiled journal on a library shelf may truly appreciate the ease with which we can now access digital articles from the comfort of our own homes or offices. However, this progress does not come without complications. In recent years, the swift growth of available research literature accessible through novel information technologies has enabled the discovery of vast amounts of information and facilitated immediate access to primary research across multiple data collections resulting in a “data deluge” [2, 15, 27].

Seemingly, this rapid access to research could be considered an asset for scoping reviews, a type of literature review designed to map the conceptual boundaries of a topic or field [23]. The ability to synthesize findings from a variety of study types (i.e., both quantitative and qualitative) has contributed to the increase in popularity of scoping reviews [18]. However, traditional scoping review methodologies struggle to meet the expanding number of relevant and available publications.

This was the challenge our team encountered when attempting to conduct a scoping review on the topic of Virtual Human (VH) realism. VHS are fundamentally computer programs, code and data designed to give the illusion of a human being [5]. In order for this illusion to be successful, the contributions of multiple disciplines (e.g., computer vision, animation, natural language processing, non-verbal behavior generation, etc.) are leveraged and harmoniously integrated into a digital representation of humanness.

Given the breadth and interdisciplinary nature of this topic, our initial scoping review resulted in a sizable body of research. The total number of resources we identified and advanced to the screening stage of the scoping review process was 34,153 scientific articles. This incredibly large dataset inspired us to investigate document-mining algorithms to facilitate and accelerate a comprehensive scoping review that would enable us to efficiently explore the last thirty years of research on the digital representation of humans.

The purpose of this case study is to define the user-centered design process our interdisciplinary team of computer scientists, librarians, and social scientists engaged in to explore the vast literature on virtual humans and present our resulting research tool: a semantically organized, interactive evidence map, clustered by

salient topics and linked to Google Scholar to expedite the discovery of relevant resources (Fig. 2). Specifically, we aimed to achieve a number of desiderata with this tool including: (1) **replicability**, (2) **objectivity**, (3) **automation & scalability**, and (4) **ease of discovery**. We employed a user-centered design process guided by the Information Systems Research (ISR) framework to define the parameters of an AI-enabled accessible research tool with an explorable user interface navigable by non-technical collaborators. To enable replicability, we provide full access to the dataset, the interactive evidence map, and source code. This work explicitly defines what was “under-the-hood” of our recent publication defining this novel semi-automated review process [19].¹

2 RELATED WORK

Scoping reviews are a relatively new approach to evidence synthesis [20]. Unlike systematic reviews that are guided by a highly focused research question focused on a particular study design, scoping reviews consolidate findings from a variety of studies, including both quantitative and qualitative, making this type of evidence synthesis particularly relevant to disciplines with emerging evidence, bodies of literature that exhibit a large, complex, or heterogeneous nature, cross-discipline investigations, or fields that typically do not conduct randomized control trials [1, 18, 22]. This broad application of scoping reviews has contributed to their increasing popularity; a single scoping review was published in the year 2000 and a reported 3,093 were published in 2019 [23, 25].

Though scoping reviews are typically conducted with broader inclusion criteria than a systematic review, scoping reviews require rigorous and transparent methods to produce valid and reliable results [20]. A recent scoping review of scoping reviews found that across disciplines the majority of published scoping reviews follow the five-step process originally outlined by Arksey and O'Malley (See Fig 3) [29]. Additionally, this scoping review revealed that the main purposes that Arksey and O'Malley first identified for conducting a scoping review remain among the most common goals of published reviews including (1) creating an evidence map, (2) identifying key themes and the breadth of the research (3) detecting gaps in the existing literature and (4) determining the feasibility of conducting a full systematic review [1, 24].

Navigating this five-stage framework takes a significant amount of time; authors have reported scoping reviews can take up to 20 months to complete [24]. Moreover, building a team that incorporates diverse expertise in the library sciences, review methods, and domain expertise for the topic under review has been shown to improve the quality of the review, but the resources and time needed to assemble these teams may be too big a task for many researchers and research budgets [11].

Even with the boon of our diverse research team, the amount of literature we identified for the VH realism project was too vast for us to manually review in a timely manner. Thus, this case study will explore how we built upon this established scoping review methodology through a user-centered design process that applied novel representation learning approaches to support review teams in successfully handling this modern-day data deluge.

¹<https://github.com/USC-ICT/VHFC>

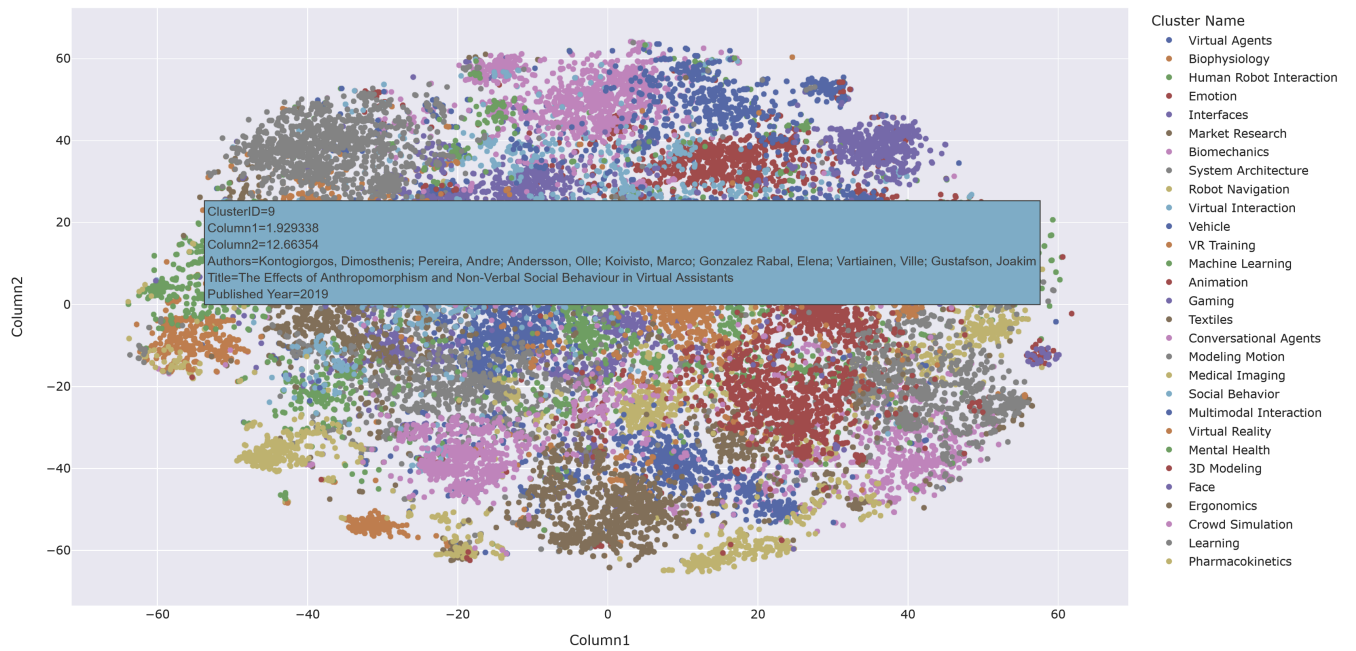


Figure 2: Visualization of the final $k = 29$ clustering for VH dataset. To access the interactive map go to: https://vhtoolkitwww.ict.usc.edu/VHFC/fidelity_clusters_google_scholar.html

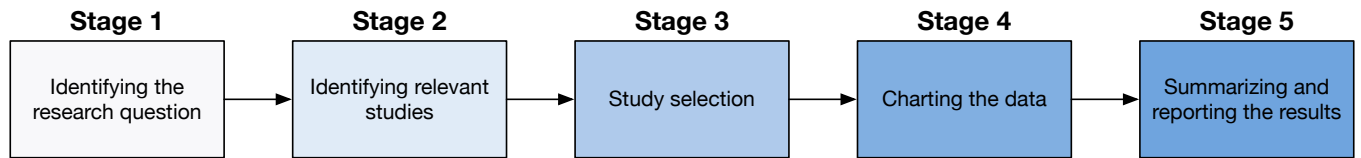


Figure 3: Scoping review stages.

3 METHODS

The date range for our VH realism scoping review spanned 1990 to 2022 to capture the complex history of VH research and seminal works on VH emotion and perception. Initial inclusion criteria for the scoping review were: journal articles, conference proceedings, dissertations and theses, and review articles published in English. A main goal of this project was to capture and catalogue all works on VH realism; an initial review revealed that the extant VH literature utilized several terms to address realism (e.g., fidelity, uncanny valley, humanness, etc.) and also works could be produced that explored aspects of realism without directly referencing realism (or synonyms) in the title, abstract or text. Therefore we chose to cast a wide net and use the search terms domain experts agreed would be most likely to capture relevant literature on VHs that could later be vetted for its relevance to realism. Furthermore, we recognized that this dataset could offer value beyond our initial inquiries into realism and could be leveraged to address further inquiries related to existing research on virtual humans.

The search terms we chose included, “virtual human”, “virtual agent”, and “embodied conversational agent” or “ECA”, popular in

academic research, and “digital human”, a term often used by industry. We collected resources from the following databases: ACM, ARXIV, EBSCO, Engineering Village, IEEE XPLORE, Gale, ProQuest, PubMed, ScienceDirect, Scopus, Wiley, and Web Of Science. Newspapers, magazines, press releases, books, book chapters, conference reviews, editorials, notes, letters, short surveys, retracted, erratum, and undefined articles were excluded. Given the breadth of the topic, the initial scoping review resulted in a sizable body of research to be synthesized. The total number of resources advancing to data screening was 34,153 in our VH research dataset after removing duplicates. This number inspired our team to investigate automating portions of the scoping review process [19].

The ISR Framework conceptualizes the design process as an embodiment of three closely related cycles of activities (Fig. 4). As applied to the design of this interdisciplinary research tool, we aimed to understand the requirements of end-users (e.g., researchers, librarians, etc.) who would conduct scoping reviews in the *Relevance Cycle*. As part of the *Rigor Cycle*, we explored the extant literature on current data-mining methods in the information sciences to connect the design science activities with the knowledge base of scientific foundations, experience, and expertise. The central *Design*

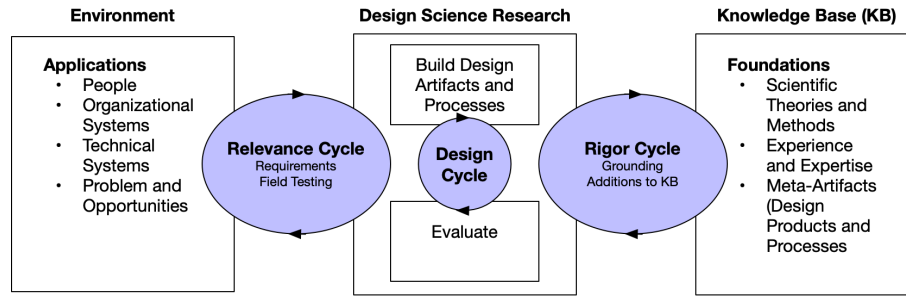


Figure 4: The Information Systems Research framework adapted from Hevner [14].

Cycle iterates between the core activities of building and evaluating the design artifacts and processes of the research. Here, we produced and refined our research tool, resulting in an interactive evidence map of all discoverable research related to VHs, organized by topic clusters, and directly linked to Google Scholar to aid in the discovery of relevant research [14].

For the purposes of this case study we will briefly describe our team activities in the overarching ISR framework. We will share our findings, lessons learned, the limitations of our current work, and future directions of this project.

The Relevance Cycle The goal of the relevance cycle was to understand the desired function of a tool to accelerate scoping reviews. We conducted a series of meetings with subject matter experts in the field of library sciences with specific expertise in scoping review methodology. Additionally, we sought input on which stages of the scoping review methodology (Fig. 2) could be responsibly automated and which parts of the process would still need a human touch. Finally, we aimed to define the data that would be available to us in order to automate at least a portion of the scoping review process.

The Rigor Cycle To understand the current state of automation in reviews, our team performed a literature review of evidence synthesis techniques to inform our design process.

The Design Cycle In the design cycle we set out to define an approach that could synthesize the extant literature based on the parameters defined in the relevance cycle utilizing available data. As the ISR framework is conceptualized as an iterative process, our interdisciplinary team of computer scientists, librarians and psychologists collaborated on the realization of a functional prototype: a semantically organized interactive evidence map. Additionally, all team members provided feedback and we evolved the initial prototype based on this feedback.

4 RESULTS

4.1 Relevance Cycle

In reviewing the five-stage process outlined by Arksey and O'Malley (see Fig. 2), our subject matter experts indicated that automation could potentially expedite *Stage 4: Charting the Data* and *Stage 5: Summarizing and Reporting Results*. Several stages were deemed by our subject matter experts to still require a human touch due to the nuance and subjectivity of these tasks: defining the overarching research question (Stage 1), defining databases of interest to

be mined for relevant resources (Stage 2) as well as defining the inclusion/exclusion criteria for research relevant to the research question (Stage 3). During this cycle, it was also determined that by Stage 3 our scoping review team would have access to the title and abstract of all relevant studies. In a smaller review, it may be of added value to pull full-texts, however, in our review of our 30,000 articles, it was not deemed reasonable that the full-text of all articles could be aggregated in a timely manner. Finally, four interdependent areas were determined to be of high priority for the tool: (1) **replicability**, (2) **objectivity**, (3) **automation & scalability**, and (4) **ease of discovery for non-computer scientists**, indicating that the team valued a easily navigable UX for the organization and exploration of relevant studies.

4.2 Rigor Cycle

A review of the literature of evidence synthesis techniques revealed that currently there is a body of research focused on automating the tedious aspects of literature reviews [3, 16, 30]. Available tools and task automation algorithms range from assisted meta-search tools [30], machine learning-based abstract screening [33, 34], and automated result synthesis tools, like RevMan-HAL [28] and PRISMA [21]. For example, the tool *abstrackr* discussed in [33] relies on a semi-automated active learning approach during which the human reviewer iteratively trains a Support Vector Machine (SVM) [8] to categorize titles and abstracts as either “relevant” or “irrelevant” for the systematic review. Here, the utilized features rely on traditional representations of documents named bag-of-words [35]. While these representations have the ability to capture topics with standardized vocabulary they are not able to represent complex relationships *between* words (e.g., word order in bag-of-words is typically ignored) [35]. While extensions to bag-of-words, such as n-grams [9] are possible, they are fundamentally limited to small $n \ll 10$, due to the combinatorial explosion of the vocabulary size. Related to our priorities defined in the *Relevance Cycle*, we found document representation algorithms relying on extremely large datasets and neural network models (e.g., transformer models) have been developed that are able to learn latent concepts and representations of language that go beyond any engineered approach [7, 32]. Such neural models are specifically capable to represent complex relationships across sentences as well as entire documents and hence hold great promise for the automated analysis of scientific research documents and seemed particular salient to our work [4, 6, 19].

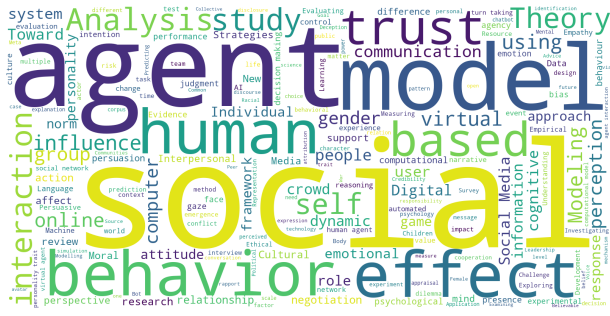


Figure 5: Cluster 19 - Social Behavior

4.3 Design Cycle

Results in the *Design Cycle* will be presented according to the two main activities defined by the ISR framework in this stage (See Fig. 4).

4.3.1 Build/Design Artifacts and Processes. Based on the findings of the *Rigor Cycle* and the priorities and available data defined in the *Relevance Cycle* (titles and abstracts), we chose to leverage document representation algorithms to identify relationships between the documents present in our scoping review dataset [19]. We devised a multi-step process visualized in Fig. 1 to visualize the complex relationships between papers, to discover emerging trends and topic clusters, as well as evidence gaps within a large unstructured document dataset.

First in this process, we leveraged the state-of-the-art document-level representation learning method SPECTER trained directly on academic paper titles and abstracts as well as their citation-relationships [7] to derive dense high-dimensional (768-dimension) numeric representations for each document. We chose SPECTER for its superior performance in the evaluation results leveraging the evaluation suite SCIDOCS [7]. For the embeddings, we feed the model paper titles and abstracts in order for the model not to be biased by author names and publication years. We did not conduct any fine-tuning on the SPECTER model as we seek to make it as broadly applicable as possible for any dataset. Second, we employed t-SNE, a dimensionality reduction algorithm, to render the high-dimensional embeddings on a two-dimensional interactive mapping [31], enabling the visual inspection of the relationships between papers. t-SNE is a non-linear dimensionality reduction technique that is widely used in machine learning [7, 10], and other fields of academic research. The goal of the t-SNE algorithm is to render similar data points close to each other and dissimilar data points further apart from each other in a low-dimensional space (i.e., two dimensions for the present work). Using t-SNE we are able to visualize the data as well as maintain complex non-linear relationships between the data points. For the purpose of this work, we leverage the SciKit Learn implementation of t-SNE with the default parameter setting and a random seed of 0 for reproducibility.²

Third, for the clustering of the documents into topics we leverage the 768-dimensional representations. We use the high-dimensional

document embeddings as input to the well-known k-means clustering algorithm [12]. To identify the optimal number of clusters k , we employ the elbow metric [26]. In our analysis, we ran the clustering for $k \in [2, 100]$. For each value of k we calculate the sum of squared errors (SSE) as the distortion score. Then we choose the elbow as the trade-off value between an optimal SSE and a small k .

Fourth, to label each of the identified clusters we employed a word-cloud algorithm to distill each cluster’s main topic in a human-readable format. Word clouds offer a quick and visually intuitive way to highlight word frequency in textual data, making them useful for initial insights in exploratory analysis. Further, word clouds have been shown to be a reliable tool for text analysis [13]. However, they have considerable shortcomings as they for example remove the context in which the words appear and focuses only on the frequency. While word clouds serve as a valuable tool for overviewing themes and deriving topics from large bodies of data, they should be complemented with more sophisticated analyses to ensure a comprehensive understanding of the data as suggested in [19]. For the purpose of this work, we use a common Python word cloud package³. Before running the algorithm we removed common words known as *stopwords* (e.g., *a*, *do*, *get*, *she*, or *I*) to render the word-cloud plots more meaningful. Specifically, we use the standard stopwords dictionary that accompanies the Python implementation of the word cloud library. Once the word clouds (cf. Fig 5) were rendered, two reviewers reviewed each plot carefully and identified keywords present in each plot to provide a meaningful label to each cluster. Labeling all 29 identified clusters took approximately 5 hours of work per reviewer.

This 4-step process enabled the creation of our research tool: a semantically organized interactive evidence map to clearly visualize and navigate the dataset in a 2-dimensional space (See Fig. 2).

4.3.2 Evaluate. After the definition of processes and creation of the research tool, we asked our multidisciplinary team to provide feedback. This research tool and associated clustering techniques were identified to have achieved two main goals of scoping reviews: (1) create an evidence map and, (2) identify key themes and breadth of research via cluster labels. Though we found this research tool expedited the organization and navigability of our massive dataset, teammates were curious if more could be done to automate the two remaining goals of a scoping review, (3) detect gaps in the existing literature and, (4) determine the feasibility of conducting a full systematic review.

To identify if a knowledge gap exists within scoping review datasets, we propose the use of keyword-matching techniques applied to the titles of the documents. To illustrate the approach we conduct a keyword-matching example for the broader topic of *military* by searching for the keywords *military*, *soldier*, *veteran*, *weapon*, *army*, *air force*, *navy*, *armor* in the dataset. We visualize the result in Fig. 6. In the *VH dataset*, research regarding the military is spread out amongst topic clusters, providing evidence that there is diffuse research across topics, however, there is a lack of thematic concentration in a single domain.

Regarding goal 4, one primary way to assess if an in-depth systematic review is feasible is “when at least ten studies are available

²<https://scikit-learn.org/stable/modules/generated/sklearn.manifold.TSNE.html>

³https://github.com/amueller/word_cloud

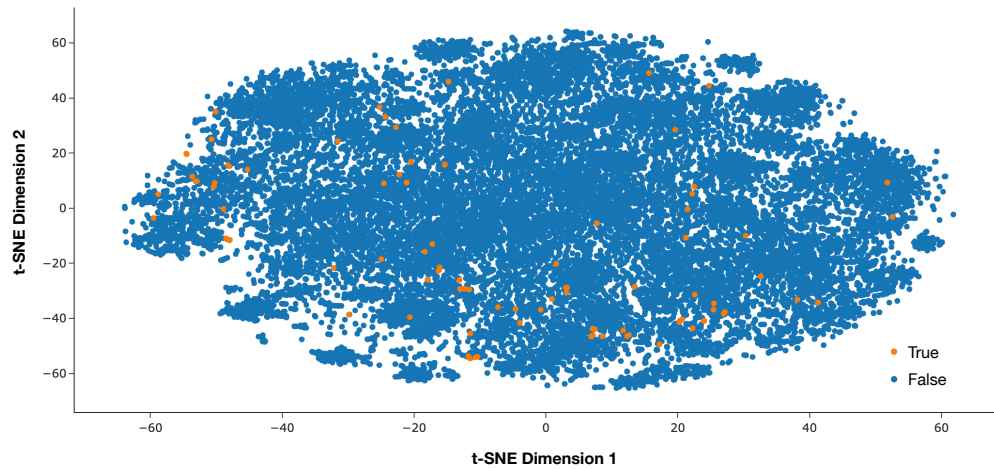


Figure 6: Visualization of the spread of the “military” topic for the virtual human dataset. The topic is identified using a set of keywords related to the military, including: *military, soldier, veteran, weapon, army, air force, navy, armor*. Matching documents in the dataset are visualized with an orange dot and those that do not are visualized with a blue dot.

on a specific topic” [29]. Leveraging the semi-automatic clustering techniques and keyword spotting we believe it is possible to evaluate whether a systematic review can be pursued. For example, based on these metrics, one can quickly ascertain that a systematic review of the use of virtual humans in the context of “military” simulation and training is feasible and could be conducted.

Additionally, though this research tool provided a navigable evidence map, the team requested additional functionality to speed up the rate of discovery. At this point in the scoping review process, the team wondered if there was a way we could use this research tool in the retrieval of full texts. To accomplish this, we linked the evidence map with Google Scholar. Clicking on any point (i.e., article) on the map will open the corresponding record for this article in Google Scholar making full-text access only a few clicks away on GitHub. This functionality leverages Google Scholar’s ability to create queries using URL parameters, which we populated with both author names and the article title.

5 DISCUSSION

Engaging in the ISR process in an attempt to solve our own “data deluge” problem when researching VHs resulted in valuable lessons in collaboration and tool development.

A multidisciplinary team was essential to solving our interdisciplinary problem: This project would not have been possible if we were not working as part of a multidisciplinary team. Alone, our technical team would not have understood the goals of a scoping review or the stage-wise manual process most scoping review teams engage in to complete their work. Without the technical expertise of the computer scientists on our team, the semi-automated process would not have been realized. Finally, without thoughtful feedback from our entire team, this tool would not have evolved to its current usable state with direct links to full-text documents.

In-house training is essential: Our computer scientists, librarians, and social scientists all reported an increase in their knowledge

in domains outside their own areas of expertise as a result of this project. Early on in the process, we dedicated meeting time to ensure all parties had a solid understanding of VHs and the scoping review process. As the project progressed, meeting time was dedicated to domain experts in machine learning who led tutorials on the underlying mechanisms of the research tool including representation learning and k-means clustering, as well as possible avenues of exploration and limitations of the tool. This dedication to education benefited the project by creating a base level of knowledge in the library sciences, VHs, and machine learning techniques across team members, which aided communication, workflow, and report writing.

The ISR Framework is a useful tool for multidisciplinary collaboration: The ISR Framework helped to guide collaboration by setting goals and establishing an environment of iteration. At the beginning of our project, the library science experts on our team led the establishment of the following desiderata for the research tool: (1) **replicability**, (2) **objectivity**, (3) **automation & scalability**, and (4) **ease of discovery**. Regarding **replicability**, we make all the data, search criteria, source code, and parameters used in the approach available through an open-source repository hosted on GitHub. Second, we leverage only open-source software and well-established algorithms. Regarding **objectivity**, we sought to reduce bias by eliminating subjective judgment from our semi-automated approach. We employed well-defined automatic steps to evaluate and verify characteristics of the learned representations, such as the number of discovered topics within a scoping review dataset and used the elbow method to identify an optimal number of k clusters for the k-means algorithm. In terms of **scalability**, we also applied this semi-automated approach to a much smaller dataset focused on personalized avatars and were able to replicate the results demonstrating that the here presented approach can be applied to datasets of any size and help automate some of the most overwhelming steps of scoping reviews [19]. Specifically, the approach was able to significantly speed up the identification of

topics, their trends, and the presence of studies covering certain topics. Finally, we provide a set of tools and approaches that render scoping review datasets more accessible and **discovery** easier. Together with the interactive evidence map that allows the user to hover over titles⁴ and explore articles related to VHs with ease as well as link to Google Scholar records and full-text access, this represents a significant step towards more accessible scoping review data discovery and full-text access.

Intentional feedback loops enable multidisciplinary input throughout the development life cycle: We envisioned feedback loops as bidirectional interactions between the technical developers and the information science end-users as well as our methodological experts and qualitative raters. We cultivated a feedback-friendly environment, scheduling weekly feedback sessions based on the developmental and methodological progress that had been made in the previous week. This tool was never and is still not considered “done” and will continue to evolve as we as a team continue to work the *Relevance, Rigor, and Design Cycles*. At its core, ISR is an iterative process, which helped remove the pressure from brainstorming and review sessions as each team member was able to provide feedback at any point (not solely during the scheduled sessions) with the goal of adapting our processes and tools through the unique perspectives of all end-users.

6 LIMITATIONS

Of course, the here presented work is not without limitations. Though our first aim is **replicability** and we make all the data, search criteria, source code, and parameters available, this step may be daunting for teams without some level of computer science knowledge. Additionally, while our semi-automated approach speeds up data synthesis, it still requires the qualitative analysis of clusters by domain experts. Moreover, the elbow method provides a mathematically optimal trade-off between the number of clusters k and the distortion score. Therefore, researchers should employ their own judgment when assessing the exact number of clusters and may need to discuss merging or splitting clusters given their observations. It should also be noted that a full manual screening approach can currently identify irrelevant articles that were introduced into the dataset during the initial search process (e.g., irrelevant articles retrieved by keyword search, tables of content, articles with missing fields, etc.) while the automated process right now does not include a strong filtering approach resulting in a data set that includes articles adjacent and highly related to the domain of the VHs. This filtering approach could be introduced in a future update of the work but was out of scope for our initial version. While we worked to eliminate bias, this remains a semi-automated approach, still vulnerable to human error. Specifically, when analyzing the topic of each cluster, we also resorted to a method that only assists human subjective judgments rather than a fully automated term recognition (ATR) algorithm [17]. This is motivated by the fact that due to the nature of the collected dataset often ATR algorithms would identify terms that are present in the search criteria of the scoping review, such as “virtual human” instead of the actual topic of the cluster within the realm of virtual human

research. We believe that these ATR methods have the potential to further refine the approach taken in this work, but this will require further iterations, subject to future investigations [19]. The review of this tool was also limited to our small team. Further evaluation of this tool is necessary and planned for the coming academic year. Last but certainly not least, in fact perhaps most importantly, while this approach controls for the subjective bias introduced by manual screening, automated methods are certainly not without their own biases. We advocate for continuous screening for novel and improved representation learning algorithms that could replace the SPECTER model leveraged in this work.

7 CONCLUSIONS

Our work reports on the partnership of social science researchers, computer scientists, and librarians in the development of a semi-automated scoping review approach to solve a problem we faced as an multidisciplinary review team. This methodology can rapidly identify literature that should be further reviewed by researchers wishing to establish the current state of knowledge in a particular field or across multiple disciplines. The application of validated semi-automated reviews will be a valuable tool to improve the efficiency of evidence synthesis projects and improve communication across disciplines as advances in information sciences inevitably increase the access to and volume of articles available to researchers.

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⁴To access the interactive map go to: https://vhtoolkitwww.ict.usc.edu/VHFC/fidelity_clusters_google_scholar.html

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